



Solution Set

Question 1

A market-risk team must reduce factor sensitivities using two hedges. Hedge x reduces equity delta by 6 and credit-spread CS01 by 1. Hedge y reduces equity delta by 2 and CS01 by 4. The desk needs at least 120 units of delta reduction and 80 units of CS01 reduction. Costs are 5 and 4 per unit. Formulate and solve the linear program.

Answer key

Let x and y be hedge units. The linear program is

$$\begin{aligned} \min \quad & 5x + 4y \\ \text{s.t.} \quad & 6x + 2y \geq 120, \quad x + 4y \geq 80, \quad x, y \geq 0 \end{aligned}$$

At an interior optimum both risk constraints often bind. Solve

$$\begin{aligned} 6x + 2y &= 120, & x + 4y &= 80 \\ 3x + y &= 60, & x + 4y &= 80 \Rightarrow x = \frac{160}{11}, \quad y = \frac{180}{11} \end{aligned}$$

The cost is

$$5 \frac{160}{11} + 4 \frac{180}{11} = \frac{1520}{11} \approx 138.18$$

Axis feasible costs are higher, so this binding intersection is optimal.

Question 2

Derive the dual of the hedge optimization problem in Question 1 and interpret the dual variables.

Answer key

The primal is a minimization with greater-than constraints, so the dual is a maximization with less-than constraints:

$$\begin{aligned} \max \quad & 120u + 80v \\ \text{s.t.} \quad & 6u + v \leq 5, \quad 2u + 4v \leq 4, \quad u, v \geq 0 \end{aligned}$$

Here u is the marginal cost of one extra required unit of equity delta reduction, and v is the marginal cost of one extra required unit of CS01 reduction, while the current active set remains unchanged.

Question 3

Use complementary slackness to solve the dual from Question 2 and verify strong duality.

Answer key

Because both primal hedges are positive in Question 1, both dual resource inequalities bind:

$$\begin{aligned} 6u + v &= 5, & 2u + 4v &= 4 \\ u &= \frac{8}{11}, & v &= \frac{7}{11} \end{aligned}$$

The dual value is

$$120 \frac{8}{11} + 80 \frac{7}{11} = \frac{1520}{11}$$

This equals the primal minimum cost, confirming strong duality for this feasible bounded LP.

Question 4

A VaR-constrained allocation chooses weights w in two strategies. Expected returns are 8 percent and 12 percent. Risk-budget usage per dollar is 2 and 5. Capital usage per dollar is 4 and 3. Maximize expected return subject to risk budget 10, capital budget 12, and long-only weights. Solve graphically by corner checking.

Answer key

The problem is

$$\begin{aligned} \max \quad & 0.08w_1 + 0.12w_2 \\ \text{s.t.} \quad & 2w_1 + 5w_2 \leq 10, \quad 4w_1 + 3w_2 \leq 12, \quad w_1, w_2 \geq 0 \end{aligned}$$

Candidate corners are

$$(0, 0), \quad (3, 0), \quad (0, 2), \quad (w_1, w_2) \text{ where both constraints bind}$$

Solving the binding pair gives

$$2w_1 + 5w_2 = 10, \quad 4w_1 + 3w_2 = 12 \Rightarrow w_1 = \frac{15}{7}, \quad w_2 = \frac{8}{7}$$

Objective values are 0, 0.24, 0.24, and

$$0.08\frac{15}{7} + 0.12\frac{8}{7} = 0.3086$$

so the intersection is optimal.

Question 5

Prove that if an LP has a finite optimum over a nonempty bounded polygon, at least one optimum is attained at an extreme point.

Answer key

Every feasible point in a bounded polygon can be written as a convex combination of extreme points:

$$x = \sum_i \lambda_i v_i, \quad \lambda_i \geq 0, \quad \sum_i \lambda_i = 1$$

For a linear objective $c^\top x$,

$$c^\top x = \sum_i \lambda_i c^\top v_i$$

A weighted average cannot exceed the largest term in the average. Therefore some extreme point has objective value at least as good as x . Taking x to be optimal proves that an extreme-point optimum exists.

Question 6

A hedge LP is infeasible after the desk adds liquidity and notional caps. Give a rigorous diagnostic checklist that a quant interviewer would expect.

Answer key

Check whether each single constraint is feasible, then add constraints one group at a time: risk-reduction constraints, sign constraints, notional caps, liquidity caps, and concentration limits. Inspect units, signs, and whether hedges reduce or add the exposure. A common failure is requiring more risk reduction than the maximum possible under notional caps:

$$\max_{x \in \mathcal{C}} a^\top x < b$$

If this inequality holds for any required exposure b , the full LP cannot be feasible.

Question 7

Formulate an L1 transaction-cost hedge that minimizes absolute residual risk exposure after trading two hedges.

Answer key

If A is the exposure matrix, h is the hedge vector, and e is current exposure, the residual is $e - Ah$. The L1 objective can be linearized with auxiliary variables u_i :

$$\begin{aligned} \min \quad & \sum_i u_i + \lambda c^\top |h| \\ & -u_i \leq e_i - (Ah)_i \leq u_i, \quad u_i \geq 0 \end{aligned}$$

If hedge notional also has absolute-value costs, introduce positive and negative parts $h = h^+ - h^-$ with $h^+, h^- \geq 0$. This is how an apparently nonlinear absolute-value problem becomes an LP.

Question 8

What is degeneracy in an LP, and why can it matter in hedge optimization interviews?

Answer key

Degeneracy means more constraints bind at a solution than are needed to determine the basic variables, or equivalently a basic variable is zero. In hedge optimization, degeneracy can mean there are multiple economically similar hedges with the same cost. It matters because small data changes can make the optimizer jump to a different hedge portfolio even though the objective value changes only slightly.

Question 9

A portfolio optimization LP has multiple optimal solutions along an edge. How would you choose a practical hedge?

Answer key

If the objective is flat along an optimal edge, add a secondary criterion that reflects execution reality: smaller gross notional, fewer instruments, lower bid-ask cost, higher liquidity, or smaller model risk. Mathematically one can solve a second LP restricted to the optimal face:

$$c^\top x = z^*, \quad x \in \mathcal{P}$$

and minimize the chosen secondary objective there.

Question 10

Explain why dual prices are local quantities. Give a market-risk example.

Answer key

A dual price is valid while the active constraints remain unchanged. If a VaR-reduction requirement is increased slightly, the shadow price estimates the incremental cost. But if the increase is large enough to hit a notional cap or make a new hedge attractive, the active set changes and the old dual price no longer describes the marginal cost.

Question 11

A desk minimizes funding cost

$$c^\top x$$

subject to

$$Ax \geq b$$

Explain the economic meaning of reduced cost for an unused funding source.

Answer key

For an unused variable x_j , reduced cost compares its actual cost to the value of the resources it provides under the dual prices:

$$\bar{c}_j = c_j - a_j^\top y$$

In a minimization problem, a positive reduced cost means the funding source is too expensive relative to the constraints it helps satisfy, so it remains unused at the optimum.

Question 12

How can a linear stress-test allocation problem become misleading even if the LP is solved exactly?

Answer key

The LP solution is exact only for its stated model. It can be misleading if exposures are nonlinear, if hedge sensitivities change under stress, if liquidity evaporates, if transaction costs are nonlinear, or if constraints are missing. A rigorous quant answer separates optimization error from model-specification error.